Expressions and operator order

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Python readiness portfolio

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Expressions and operator order before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Values variables and types

Without opening the old lesson, explain one key idea from Values variables and types. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Expressions and operator order trace and record the first step you can explain without help.

Use W2: Predict then test **expression**s for **Expression**s and **operator** order. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Expressions and operator order explanation against the supplied facts before discussing it.

Use the worked case for **Expression**s and **operator** order. Proposed explanation: “All operations are read left to right regardless of **precedence**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Group an expression deliberately

> Predict each line. Explain which calculation groups addition before division. What would (17+5)%5 return, and how do /, // and % differ for the supplied positive integers?

```
print((17 + 5) // 5)
print(17 + 5 // 5)
print(17 / 5, 17 // 5, 17 % 5)
```

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 State the intended grouping

> State the intended grouping. Apply the supplied Expressions and operator order facts and keep your reasoning.

Write an **expression** for three packs of eight counters plus two loose counters. Explain why adding two within the **parentheses** for every pack describes a different situation.

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Predict then test expressions

> Predict then test expressions. Apply the supplied Expressions and operator order facts and keep your reasoning.

Predict 16 - 5 - 3, 16 - (5 - 3), and 18 / 3 * 2. Write the grouping, run each line and record actual results separately.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Explain whole groups

> Explain whole groups. Apply the supplied Expressions and operator order facts and keep your reasoning.

For 17 counters in groups of 5, explain /, // and %. Show a multiplication-and-addition check for the complete groups and **remainder**.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Translate a new situation independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Translate a new situation independently. Apply the supplied Expressions and operator order facts and keep your reasoning.

Four trays hold six tiles each and five tiles are loose. Write and evaluate the total **expression**. Then pack the total into complete groups of four and find the **remainder**. Explain why 4 * (6 + 5) is wrong for this story.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Check a limit

> Check a limit. Apply the supplied Expressions and operator order facts and keep your reasoning.

Explain why a group size of zero must be excluded. State the assumptions that let complete groups and a **remainder** describe the packing task.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Expressions and operator order to Python readiness portfolio, identify a limit and choose the next useful test.

Add a labelled **expression**s and **operator** order **evidence** sheet to Python readiness portfolio. Show how you can evaluate numeric **expression**s using python's grouping rules.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

Pictorial case practice: Expressions and operator order

> Parentheses change an expression. A toy shelf calculator compares adding a fixed cost after multiplication with multiplying an entire group. Explain the labelled step before checking the changed condition.

Parentheses change an **expression**. A toy shelf calculator compares adding a fixed cost after multiplication with multiplying an entire group.

Python 3: print(3+2*5); print((3+2)*5). **Input**s are dimensionless for this arithmetic exercise.

AI ACADEMY / PYTHON-BRIDGE/3

Expressions and operator order

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Use precedence

In 3+2*5, multiplication happens first.

↓



2. Compute first result

2*5 is 10; adding 3 gives 13.

↓



3. Compute grouped result

Parentheses give 3+2=5 first; multiplying by 5 gives 25.

CHANGE ONE THING

Replace the final multiplier 5 with 4 in both expressions.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Trace the operation order in both **expression**s. Identify which picture step changes with **parentheses** and predict both results when the multiplier becomes four.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation.

Use actual Python 3 results to check the **prediction** after reasoning first.

WORKBOOK · 13 OF 15

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Python readiness portfolio record.

Project: Python readiness portfolio

Current milestone: Python readiness portfolio. Keep runnable code, **prediction**s, actual **output**s, one repaired error and an explanation of a function on changed **input**s.

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Parentheses change an **expression**. A toy shelf calculator compares adding a fixed cost after multiplication with multiplying an entire group.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? **Parentheses** give 3+2=5 first; multiplying by 5 gives 25.

1. Compute grouped result

2. Use **precedence**

3. Compute first result

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Compute grouped result **Parentheses** give 3+2=5 first; multiplying by 5 gives 25. **Parentheses** explicitly group an operation.

Without them, Python multiplication takes priority over addition; the same numerals can therefore describe different procedures.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Replace the final multiplier 5 with 4 in both **expression**s. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. The first result is 11; the grouped result is 20.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The first result is 11; the grouped result is 20. 3+2*4=3+8; (3+2)*4=5*4.

Repair a missing building block

Review Values variables and types (python-bridge/2). Goal: Distinguish integer, floating-point, text and Boolean values. Short practice: Predict both print **output**s and explain why they differ.

Does count+tag work, and what happens to each valid **output** after count becomes ten?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

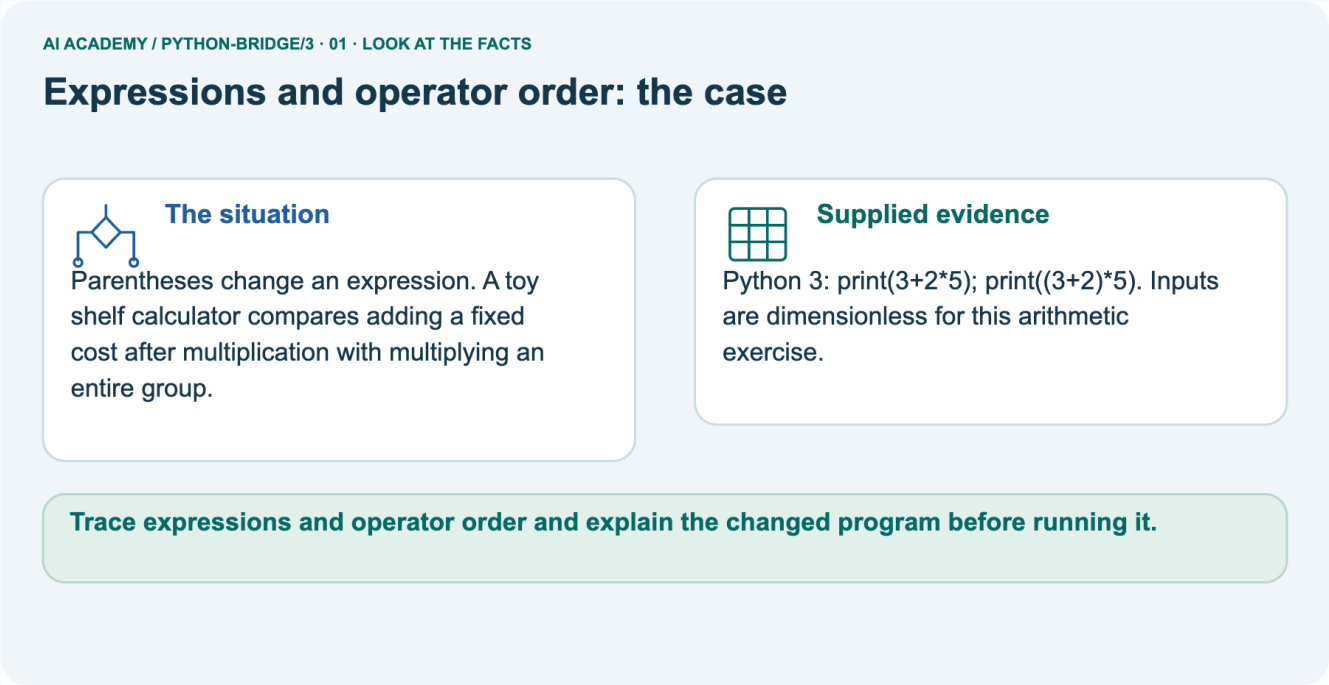
> **What I repaired, the evidence I kept and the next test**
>
> | | |
> |---|---|
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WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Expressions and operator order picture from its labels, then record a relevant limit.

01. Expressions and operator order: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

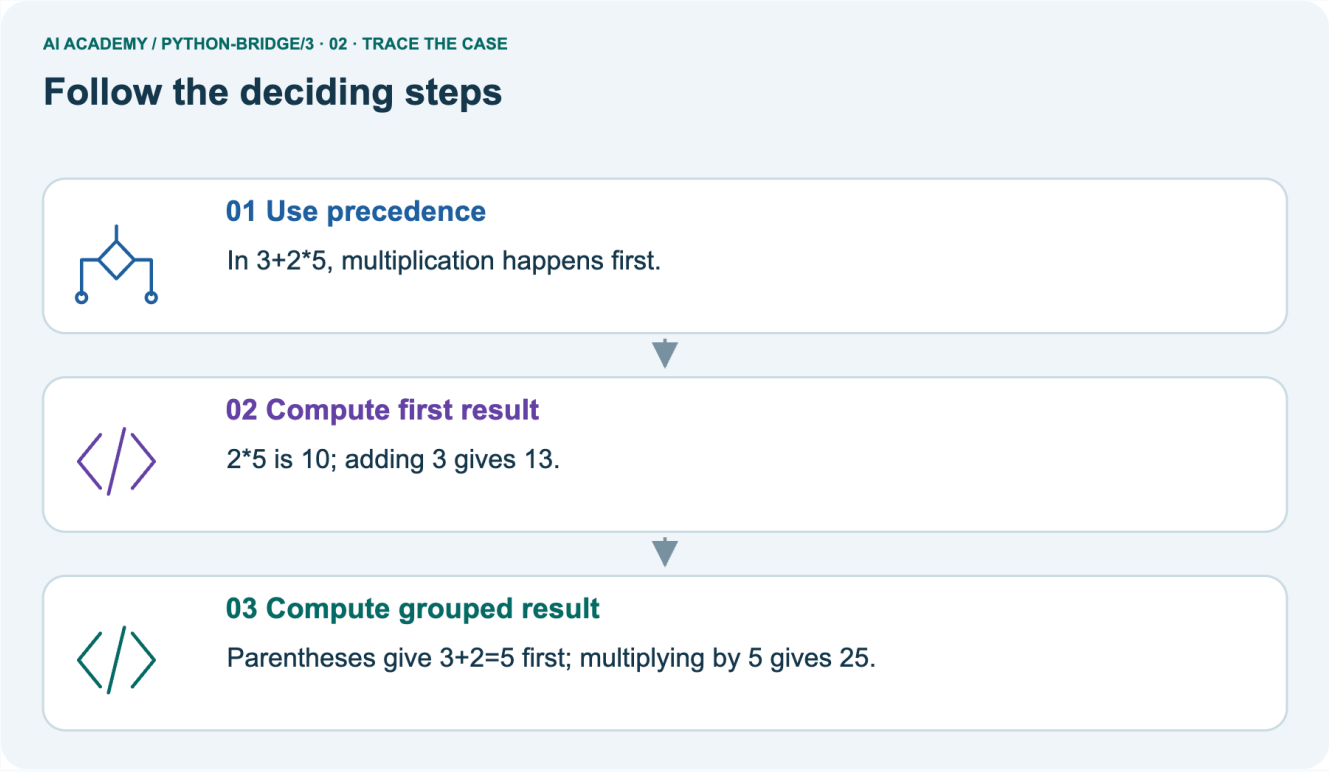
Parentheses change an **expression**. A toy shelf calculator compares adding a fixed cost after multiplication with multiplying an entire group. Supplied facts: Python 3: print(3+2*5); print((3+2)*5).

Inputs are dimensionless for this arithmetic exercise.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Use **precedence**: In 3+2*5, multiplication happens first. → Compute first result: 2*5 is 10; adding 3 gives 13. → Compute grouped result: **Parentheses** give 3+2=5 first; multiplying by 5 gives 25.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Two expressions with different meanings

AI ACADEMY / PYTHON-BRIDGE/3 · 03 · INSPECT THE LESSON MODEL

Two expressions with different meanings

Expression	First calculation	Final value
3 * 8 + 2	3 * 8 is 24	24 + 2 is 26
3 * (8 + 2)	8 + 2 is 10	3 * 10 is 30
(3 * 8) + 2	3 * 8 is 24	24 + 2 is 26

Adding two inside every pack differs from adding two loose counters once.

Follow the labelled picture with the matching explanation. Open the vector diagram

Two **expression**s with different meanings. 3 * 8 + 2 | 3 * 8 is 24 | 24 + 2 is 26. 3 * (8 + 2) | 8 + 2 is 10 | 3 * 10 is 30. (3 * 8) + 2 | 3 * 8 is 24 | 24 + 2 is 26

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.

AI ACADEMY / PYTHON-BRIDGE/3 · 04 · COMPARE THE CASES

Change one thing. Trace what follows.

Original case

Parentheses give 3+2=5 first; multiplying by 5 gives 25.

Changed condition

Replace the final multiplier 5 with 4 in both expressions.

Worked changed-case result

The first result is 11; the grouped result is 20.
Why: 3+2*4=3+8; (3+2)*4=5*4.

Follow the labelled picture with the matching explanation. Open the vector diagram

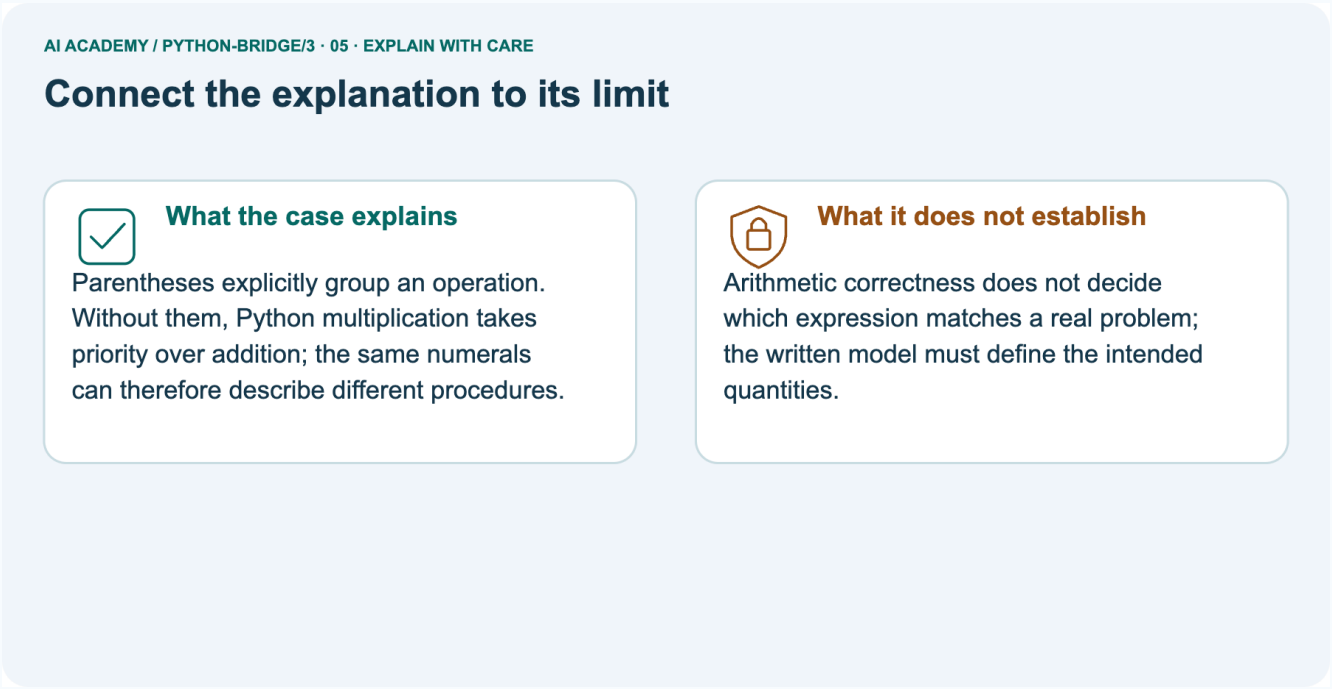
Original case: **Parentheses** give 3+2=5 first; multiplying by 5 gives 25. Changed condition: Replace the final multiplier 5 with 4 in both **expression**s. Expected result: The first result is 11; the grouped result is 20.

Reason: 3+2*4=3+8; (3+2)*4=5*4.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

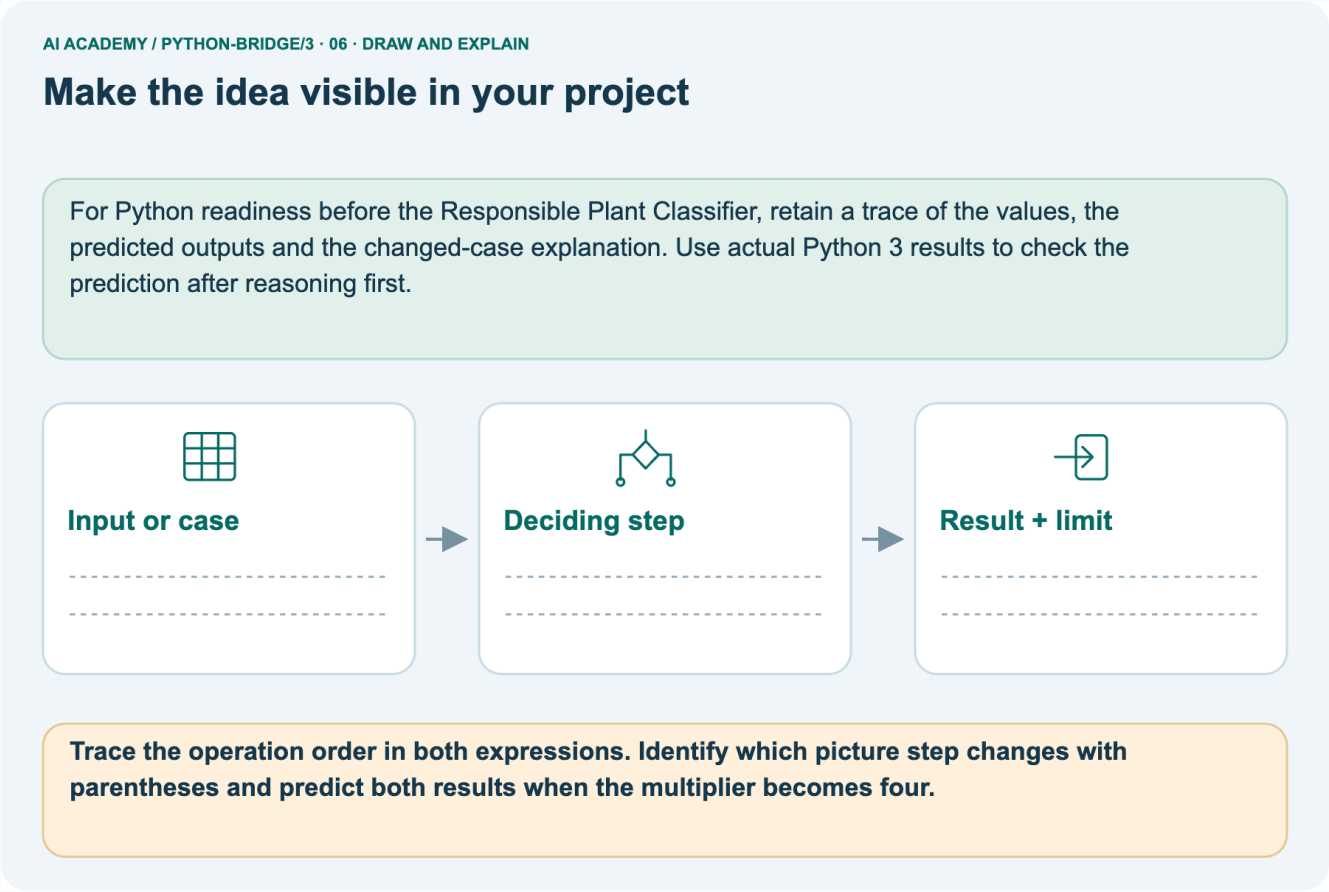
Parentheses explicitly group an operation. Without them, Python multiplication takes priority over addition; the same numerals can therefore describe different procedures.

Limit: Arithmetic correctness does not decide which **expression** matches a real problem; the written model must define the intended quantities.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Draw your **input**, deciding step, result and limit. Supported question: Trace the operation order in both **expression**s.

Identify which picture step changes with **parentheses** and predict both results when the multiplier becomes four.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.